CBC Digital Learning Platform

Stage F — Database & Data Design

ERD · Relational Schema · Data Dictionary · Sync Strategy

Stage F: Database & Data Design ← Current

1. Introduction

1.1 Purpose of Stage F

Stage F of the Unified Software Architectural Process — Database & Data Design — derives the full relational database schema from the domain model produced in Stage C. Where Stage C identified the domain entities and their conceptual relationships, Stage F translates those entities into concrete database tables, columns, data types, constraints, indices, and foreign key relationships suitable for implementation in PostgreSQL on AWS RDS.

This document constitutes the complete Data Design deliverable for the CBC Digital Learning Platform. It includes the Entity Relationship Diagram (ERD) in crow's foot notation, the full relational schema with a data dictionary for every table, and the data storage and offline synchronisation strategy.

1.2 Inheritance Pattern Decision

The User hierarchy (User → Learner, Teacher, MoESOfficial) and Content hierarchy (Content → Lesson, TeacherMaterial) are implemented using Joined Table Inheritance. This means a base table (users, content) holds all shared columns, and each subclass has its own extension table with a primary key that is simultaneously a foreign key back to the base table. This pattern:

- Eliminates NULL columns that would appear in Single Table Inheritance
- Aligns exactly with Django's multi-table inheritance ORM, requiring no custom mapping
- Keeps each table's responsibility clear and query-able independently
- Maintains referential integrity at the database level

1.3 Design Conventions

- All primary keys use UUID (gen_random_uuid()) — avoids integer sequence exposure in URLs and supports future distributed sharding
- All timestamps use TIMESTAMPTZ (timestamp with time zone) stored in UTC — correct for a Uganda-deployed platform accessed from multiple time zones
- Soft deletes: sensitive tables (users, posts) use is_active / is_deleted flags rather than physical deletion, preserving audit trails
- All ENUM types are defined as PostgreSQL custom ENUM types for type safety
- Junction tables use composite primary keys (both FKs combined) to enforce uniqueness at the DB level
- Indices are defined on all foreign keys and all frequently-filtered columns per the query patterns in the SRS

2. Entity Relationship Diagram

2.1 ERD — Crow's Foot Notation

Figure F-01 presents the complete Entity Relationship Diagram for the CBC Digital Learning Platform. The diagram uses crow's foot notation: a single perpendicular bar denotes "one", double bars denote "one and only one" (mandatory), and the crow's foot (three-pronged fork) denotes "many". Colour bands group tables by domain layer.

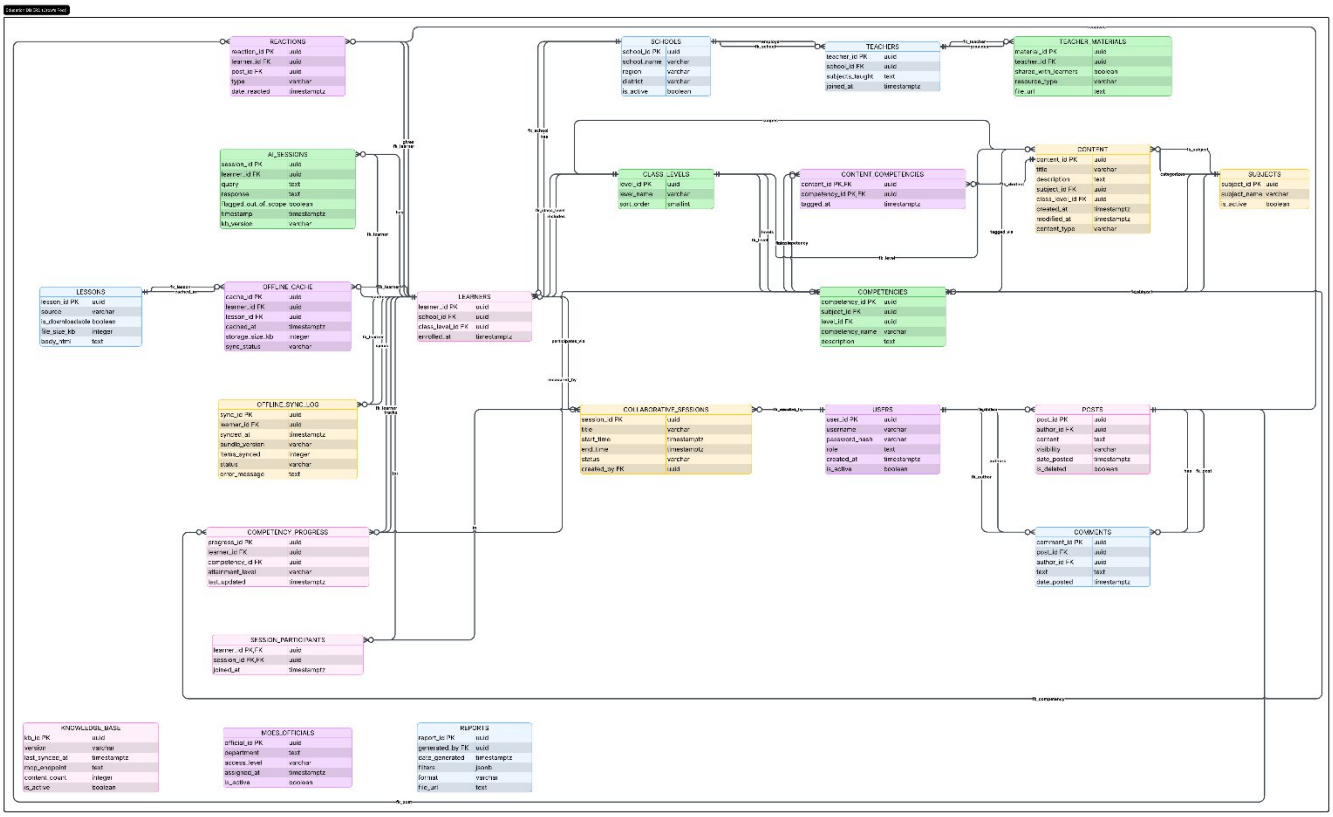


Figure F-01: CBC Digital Learning Platform — Entity Relationship Diagram (Crow's Foot Notation, PostgreSQL)

2.2 Table Inventory

The schema comprises 21 tables across five groups:

Group	Tables	Count
User Hierarchy	users, learners, teachers, moes_officials	4
Content Hierarchy + CBC Structure	content, lessons, teacher_materials, subjects, class_levels, competencies	6
Learner Interaction	posts, comments, reactions, collaborative_sessions, competency_progress, ai_sessions, offline_cache	7
AI / MoES / Supporting	knowledge_base, reports, schools	3
Junction + Sync	session_participants, content_competencies, offline_sync_log	3

3. Relational Schema & Data Dictionary

Each table below shows the full column specification, data type, constraints, index, and a description of every column. PK = Primary Key. FK = Foreign Key. UQ = Unique constraint. NN = NOT NULL.

3.1 User Hierarchy Tables

Table: users

users				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
user_id	UUID	PK, NN	PRIMARY	System-wide unique identifier for every account
username	VARCHAR(50)	NN, UQ	UQ	Unique login name chosen at registration
password_hash	VARCHAR(255)	NN	—	bcrypt-hashed password. Plain text never stored (NFR-SEC-03)
role	ENUM	NN	IDX	Values: LEARNER, TEACHER, ADMIN. Controls routing and permissions
created_at	TIMESTAMPTZ	NN, DEFAULT NOW()	—	Timestamp of account creation (UTC)
is_active	BOOLEAN	NN, DEFAULT TRUE	IDX	Soft-delete flag. FALSE = account deactivated, not deleted

Table: learners

learners				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
learner_id	UUID	PK, FK → users.user_id	PRIMARY	1:1 with users. Joined Table Inheritance extension
school_id	UUID	FK → schools.school_id, NN	IDX	Learner's registered school
class_level	ENUM	NN	IDX	Values: S1, S2, S3, S4, S5, S6. Current year group
enrolled_at	TIMESTAMPTZ	NN, DEFAULT NOW()	—	Date learner was registered on the platform

Table: teachers

teachers				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
teacher_id	UUID	PK, FK → users.user_id	PRIMARY	1:1 with users. Joined Table Inheritance

teachers				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
				extension
school_id	UUID	FK → schools.school_id, NN	IDX	Teacher's registered school
subjects_taught	TEXT[]	—	—	PostgreSQL array of CBC subject names the teacher covers
joined_at	TIMESTAMPTZ	NN, DEFAULT NOW()	—	Date teacher was registered on the platform

Table: moes_officials

moes_officials				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
official_id	UUID	PK, FK → users.user_id	PRIMARY	1:1 with users. Joined Table Inheritance extension
department	VARCHAR(100)	NN	—	Ministry department or NCDC division name
access_level	ENUM	NN	IDX	Values: STANDARD, SENIOR. Controls analytics and content access tiers
assigned_at	TIMESTAMPTZ	NN, DEFAULT NOW()	—	Date official was provisioned on the platform
is_active	BOOLEAN	NN, DEFAULT TRUE	IDX	Soft-delete flag for official accounts

3.2 Content Hierarchy + CBC Structure Tables

Table: content

content				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
content_id	UUID	PK, NN	PRIMARY	Unique identifier for all content items (base table)
title	VARCHAR(255)	NN	IDX	Display title of the content item
description	TEXT	—	—	Brief summary of the learning objective
subject_id	UUID	FK → subjects.subject_id, NN	IDX	CBC subject this content belongs to
class_level_id	UUID	FK → class_levels.level_id, NN	IDX	Target class level (S1–S6)
created_at	TIMESTAMPTZ	NN, DEFAULT NOW()	—	Creation timestamp (UTC)
modified_at	TIMESTAMPTZ	NN, DEFAULT NOW()	—	Last modification timestamp (UTC). Updated on every edit
content_type	ENUM	NN	IDX	Values: LESSON, MATERIAL. Discriminator for joined inheritance

Table: lessons

lessons				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
lesson_id	UUID	PK, FK → content.content_id	PRIMARY	1:1 with content. Extension table for NCDC lessons
source	VARCHAR(50)	NN, DEFAULT 'NCDC'	—	Content authority. Always NCDC for official curriculum lessons
is_downloadable	BOOLEAN	NN, DEFAULT TRUE	IDX	Whether this lesson may be cached for offline access
file_size_kb	INTEGER	—	—	Compressed size in KB. Used by OfflineSyncService for storage management
body_html	TEXT	—	—	Full lesson content in sanitised HTML for browser rendering

Table: teacher_materials

teacher_materials				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
material_id	UUID	PK, FK → content.content_id	PRIMARY	1:1 with content. Extension table for teacher uploads
teacher_id	UUID	FK → teachers.teacher_id, NN	IDX	Authoring teacher
shared_with_learners	BOOLEAN	NN, DEFAULT FALSE	IDX	Controls visibility to learners. FALSE = teacher-only view
resource_type	VARCHAR(80)	—	—	Descriptor: study guide, past paper, notes, worksheet, etc.
file_url	TEXT	—	—	S3 presigned URL or path to uploaded file

Table: subjects

subjects				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
subject_id	UUID	PK, NN	PRIMARY	Unique identifier for each CBC subject
subject_name	VARCHAR(100)	NN, UQ	UQ	Official CBC subject name (e.g. Mathematics, Biology, History)
is_active	BOOLEAN	NN, DEFAULT TRUE	IDX	Soft flag. FALSE removes subject from learner and teacher views

Table: class_levels

class_levels				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
level_id	UUID	PK, NN	PRIMARY	Unique identifier for each class level
level_name	ENUM	NN, UQ	UQ	Values: S1, S2, S3, S4, S5, S6
sort_order	SMALLINT	NN	IDX	Integer 1–6 for ordering levels in UI and reports

Table: competencies

competencies				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
competency_id	UUID	PK, NN	PRIMARY	Unique identifier for

competencies				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
				each CBC competency
subject_id	UUID	FK → subjects.subject_id, NN	IDX	Subject that owns this competency (composition)
level_id	UUID	FK → class_levels.level_id, NN	IDX	Class level this competency targets
competency_name	VARCHAR(200)	NN	IDX	Official CBC competency name
description	TEXT	—	—	Detailed description of the expected learning outcome

3.3 Learner Interaction Tables

Table: posts

posts				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
post_id	UUID	PK, NN	PRIMARY	Unique identifier for each learner post
author_id	UUID	FK → learners.learner_id, NN	IDX	Learner who created the post
content	TEXT	NN	—	Body text of the insight, idea, or research contribution
visibility	ENUM	NN, DEFAULT 'PRIVATE'	IDX	Values: PRIVATE, PEERS, TEACHERS, PUBLIC (NFR-SEC-05)
date_posted	TIMESTAMPTZ	NN, DEFAULT NOW()	IDX	Creation timestamp. Indexed for timeline sorting
is_deleted	BOOLEAN	NN, DEFAULT FALSE	IDX	Soft-delete flag. Preserves content for moderation audit

Table: comments

comments				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
comment_id	UUID	PK, NN	PRIMARY	Unique identifier for each comment
post_id	UUID	FK → posts.post_id, NN	IDX	Post this comment belongs to (composition child)
author_id	UUID	FK → learners.learner_id, NN	IDX	Learner who wrote the comment
text	TEXT	NN	—	Body text of the comment
date_posted	TIMESTAMPTZ	NN, DEFAULT NOW()	IDX	Creation timestamp

Table: reactions

reactions				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
reaction_id	UUID	PK, NN	PRIMARY	Unique identifier for each reaction
learner_id	UUID	FK → learners.learner_id, NN	IDX	Learner who made the reaction

reactions				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
post_id	UUID	FK → posts.post_id, NN	IDX	Post being reacted to
type	ENUM	NN	IDX	Values: LIKE, SHARE
date_reacted	TIMESTAMPTZ	NN, DEFAULT NOW()	—	Reaction timestamp

Note: A composite UNIQUE constraint on (learner_id, post_id, type) prevents duplicate reactions of the same type by the same learner on the same post.

Table: collaborative_sessions

collaborative_sessions				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
session_id	UUID	PK, NN	PRIMARY	Unique identifier for each collaborative session
title	VARCHAR(200)	NN	—	Display name of the session
start_time	TIMESTAMPTZ	NN	IDX	Scheduled or actual start time
end_time	TIMESTAMPTZ	—	—	Actual end time. NULL while session is ACTIVE
status	ENUM	NN, DEFAULT 'ACTIVE'	IDX	Values: ACTIVE, CLOSED
created_by	UUID	FK → learners.learner_id, NN	IDX	Learner who initiated the session

Table: competency_progress

competency_progress				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
progress_id	UUID	PK, NN	PRIMARY	Unique identifier for each progress record
learner_id	UUID	FK → learners.learner_id, NN	IDX	Owning learner (composition child)
competency_id	UUID	FK → competencies.competency_id, NN	IDX	Competency being tracked
attainment_level	ENUM	NN, DEFAULT 'NOT_STARTED'	IDX	Values: NOT_STARTED, IN_PROGRESS, ATTAINED
last_updated	TIMESTAMPTZ	NN, DEFAULT NOW()	—	Timestamp of last

competency_progress				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
				attainment update

Note: A composite UNIQUE constraint on (learner_id, competency_id) ensures each learner has exactly one progress record per competency.

Table: ai_sessions

ai_sessions				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
session_id	UUID	PK, NN	PRIMARY	Unique identifier for each AI interaction
learner_id	UUID	FK → learners.learner_id, NN	IDX	Learner who initiated the query
query	TEXT	NN	—	The academic question submitted by the learner
response	TEXT	—	—	The AI Tutor's contextualised response. NULL if flagged out of scope
flagged_out_of_scope	BOOLEAN	NN, DEFAULT FALSE	IDX	TRUE if the query fell outside the knowledge base (FR-AI-05)
timestamp	TIMESTAMPTZ	NN, DEFAULT NOW()	IDX	Interaction timestamp. Indexed for session history queries
kb_version	VARCHAR(20)	—	—	Version of knowledge base used to answer this query. Audit trail

Table: offline_cache

offline_cache				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
cache_id	UUID	PK, NN	PRIMARY	Unique identifier for each cache record
learner_id	UUID	FK → learners.learner_id, NN	IDX	Learner who owns this cached lesson
lesson_id	UUID	FK → lessons.lesson_id, NN	IDX	Lesson that has been cached
cached_at	TIMESTAMPTZ	NN, DEFAULT NOW()	IDX	When the lesson was downloaded for offline

offline_cache				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
				use
storage_size_kb	INTEGER	—	—	Size of the cached bundle on the device (KB)
sync_status	ENUM	NN, DEFAULT 'SYNCED'	IDX	Values: PENDING (download in progress), SYNCED (complete)

3.4 AI / MoES / Supporting Tables

Table: knowledge_base

knowledge_base				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
kb_id	UUID	PK, NN	PRIMARY	Unique identifier for the knowledge base instance record
version	VARCHAR(20)	NN	—	Semantic version of the knowledge base content (e.g. 1.0.3)
last_synced_at	TIMESTAMPTZ	NN	IDX	Timestamp of the most recent MCP content synchronisation
mcp_endpoint	TEXT	NN	—	URL of the MCP server endpoint used by the AI Tutor service
content_count	INTEGER	NN, DEFAULT 0	—	Total number of content records indexed in the knowledge base
is_active	BOOLEAN	NN, DEFAULT TRUE	IDX	Only one record should be active at a time

Table: reports

reports				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
report_id	UUID	PK, NN	PRIMARY	Unique identifier for each generated report
generated_by	UUID	FK → moes_officials.official_id, NN	IDX	MoES/NCDC official who generated the report
date_generated	TIMESTAMPTZ	NN, DEFAULT NOW()	IDX	Generation timestamp
filters	JSONB	—	—	Serialised filter parameters: date range, region, subject, school
format	ENUM	NN	—	Values: PDF, CSV. Export format selected at generation time
file_url	TEXT	—	—	S3 presigned URL for download. NULL until Celery task completes

Table: schools

schools				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
school_id	UUID	PK, NN	PRIMARY	Unique identifier for each registered school
school_name	VARCHAR(200)	NN, UQ	UQ	Official school name
region	VARCHAR(100)	NN	IDX	Uganda administrative region (e.g. Western, Northern)
district	VARCHAR(100)	NN	IDX	Uganda district. Used for analytics regional filtering (FR-M-03)
is_active	BOOLEAN	NN, DEFAULT TRUE	IDX	Soft flag. Inactive schools are hidden from registration

3.5 Junction & Sync Tables

Table: session_participants

session_participants				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
learner_id	UUID	PK (composite), FK → learners.learner_id	COMPOSITE PK	Learner participating in the session
session_id	UUID	PK (composite), FK → collaborative_sessions.session_id	COMPOSITE PK	Session the learner is participating in
joined_at	TIMESTAMPTZ	NN, DEFAULT NOW()	—	Timestamp when the learner joined the session

Table: content_competencies

content_competencies				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
content_id	UUID	PK (composite), FK → content.content_id	COMPOSITE PK	Content item being tagged
competency_id	UUID	PK (composite), FK → competencies.competency_id	COMPOSITE PK	Competency being associated with the content
tagged_at	TIMESTAMPTZ	NN, DEFAULT NOW()	—	When the competency tag was applied to this content

Table: offline_sync_log

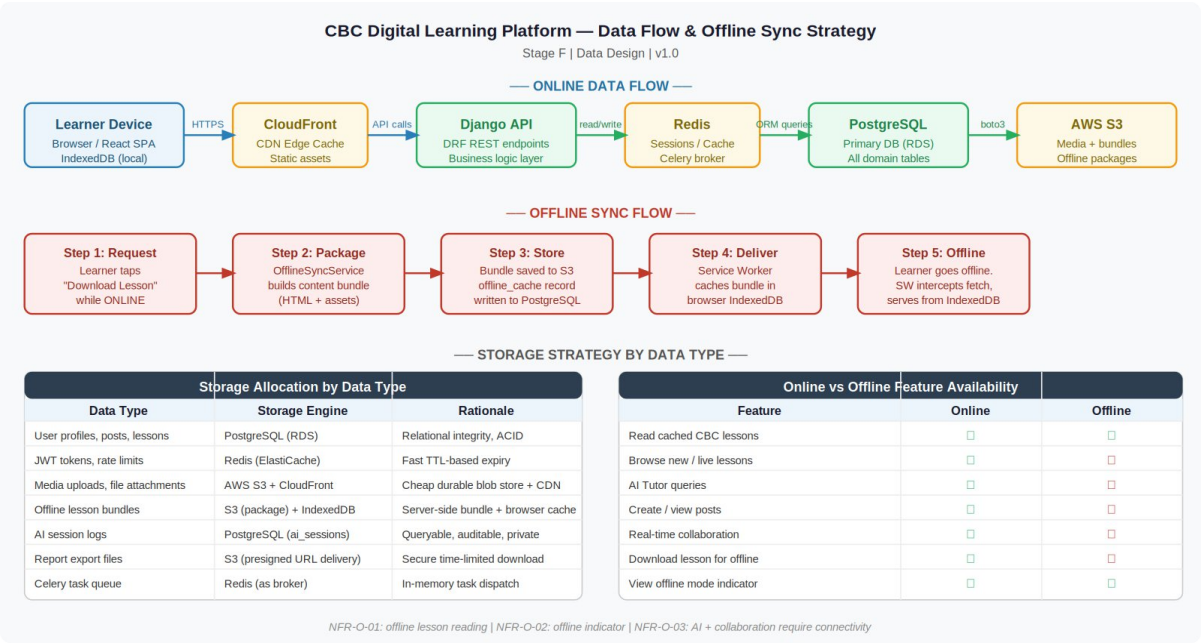
offline_sync_log				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
sync_id	UUID	PK, NN	PRIMARY	Unique identifier for each sync operation log entry
learner_id	UUID	FK → learners.learner_id, NN	IDX	Learner whose device performed the sync
synced_at	TIMESTAMPTZ	NN, DEFAULT NOW()	IDX	Timestamp of the sync attempt
bundle_version	VARCHAR(20)	NN	—	Version of the content bundle that was synced
items_synced	INTEGER	NN, DEFAULT 0	—	Number of lesson items synced in this operation

offline_sync_log				
Column Name	**Data Type**	**Constraints**	**Index**	**Description**
status	ENUM	NN	IDX	Values: SUCCESS, FAILED. Enables retry logic in OfflineSyncService
error_message	TEXT	—	—	NULL on SUCCESS. Error detail on FAILED for debugging

4. Data Flow & Offline Sync Strategy

4.1 Overview Diagram

Figure F-02 presents the complete data flow for both online and offline paths, the five-step offline sync process, the storage allocation strategy by data type, and the feature availability matrix per connectivity state.



Storage Allocation by Data Type

Data Type	Storage Engine	Rationale
User profiles, posts, lessons	PostgreSQL (RDS)	Relational integrity, ACID
JWT tokens, rate limits	Redis (ElastiCache)	Fast TTL-based expiry
Media uploads, file attachments	AWS S3 + CloudFront	Cheap durable blob store + CDN
Offline lesson bundles	S3 (package) + IndexedDB	Server-side bundle + browser cache
AI session logs	PostgreSQL (ai_sessions)	Queryable, auditable, private
Report export files	S3 (presigned URL delivery)	Secure time-limited download
Celery task queue	Redis (as broker)	In-memory task dispatch

Online vs Offline Feature Availability

Feature	Online	Offline
Read cached CBC lessons	☐	☐
Browse new / live lessons	☐	☐
AI Tutor queries	☐	☐
Create / view posts	☐	☐
Real-time collaboration	☐	☐
Download lesson for offline	☐	☐
View offline mode indicator	☐	☐

Figure F-02: CBC Digital Learning Platform — Data Flow & Offline Sync Strategy

4.2 Online Data Flow

In normal connected operation, the learner's React SPA communicates with the Django API over HTTPS via CloudFront and the Application Load Balancer. The Django backend reads and writes to PostgreSQL via the ORM for all persistent domain data, reads from and writes to Redis for session tokens, Celery task brokering, and WebSocket channel state, and reads from and writes to S3 for media files and generated reports. All static assets and the SPA build are served from S3 via CloudFront edge nodes, minimising latency for learners in Uganda relative to the AWS af-south-1 origin.

4.3 Offline Sync Flow

The offline sync process is a five-step sequence initiated by the learner while connected:

- Step 1 — Request: The learner taps "Download Lesson" in the LearnerApp UI while online. The request hits /api/v1/content/offline-bundle/ with the lesson ID.
- Step 2 — Package: The OfflineSyncService in Django retrieves the lesson body, associated assets, and metadata from PostgreSQL and S3, and packages them into a self-contained HTML bundle.
- Step 3 — Store: The bundle is saved to the S3 offline bundles prefix. An offline_cache record is written to PostgreSQL with sync_status = SYNCED. An offline_sync_log entry is created.
- Step 4 — Deliver: The bundle URL is returned to the React SPA. The PWA Service Worker intercepts the response and stores the bundle in browser IndexedDB.

- Step 5 — Offline access: When the learner's device loses connectivity, the Service Worker intercepts all fetch requests for cached lessons and serves directly from IndexedDB. The UI displays an offline mode indicator (NFR-O-02). AI tutor, posting, and collaboration features are disabled with clear messaging (NFR-O-03).

4.4 Storage Strategy

Data Type	Storage Engine	Access Pattern	SRS Ref
User profiles, posts, lessons, progress	PostgreSQL RDS	CRUD via Django ORM, ACID transactions	FR-L-01 to FR-L-09
JWT tokens, refresh token blacklist	Redis ElastiCache	TTL-based key expiry (access: 5 min, refresh: 7 days)	NFR-SEC-03, NFR-SEC-04
Rate limiting counters	Redis ElastiCache	Atomic increment with sliding window TTL	NFR-S-01
WebSocket channel state	Redis ElastiCache	Django Channels pub/sub layer	FR-L-08
Celery task queue	Redis ElastiCache (as broker)	In-memory task dispatch, result backend PostgreSQL	NFR-A-01
Uploaded media (teacher materials)	AWS S3 + CloudFront	Presigned URL upload; CDN delivery to learners	FR-T-02, NFR-P-03
Offline lesson content bundles	S3 (server) + IndexedDB (browser)	Built on request; browser service worker caches	FR-L-04, NFR-O-01
Report export files (PDF, CSV)	AWS S3	Celery-async generation; presigned URL download	FR-M-04
AI session logs	PostgreSQL ai_sessions table	Append-only writes; private read by owning learner	FR-AI-01, NFR-SEC-05
NCDC knowledge base content	PostgreSQL (via MCP Server)	Read-only by AI layer; updated by MoES/NCDC admin	FR-AI-02, FR-AI-03
React SPA static build	AWS S3 + CloudFront	Deployed once per release; CDN cached globally	NFR-P-01, NFR-AC-01

5. Index Strategy

Beyond primary key indices, the following additional indices are defined to support the platform's most frequent query patterns as derived from the SRS requirements and the API endpoint contracts in Stage E:

Table	Index Columns	Query Pattern Served
users	username	Login lookup by username (FR-L-02)
users	role	Role-based access filtering (NFR-SEC-04)
learners	school_id	School-scoped analytics queries (FR-M-03)
content	subject_id, class_level_id	CBC-filtered content listing (FR-L-03)
content	content_type	Discriminator filter for joined inheritance queries
lessons	is_downloadable	Offline-eligible lesson listing (FR-L-04)
teacher_materials	teacher_id, shared_with_learners	Teacher material feed for learners (FR-T-04)
posts	author_id, date_posted	Learner post timeline (FR-L-07)
posts	visibility, date_posted	Peer/public post feed queries (FR-L-07, FR-L-11)
competency_progress	learner_id, attainment_level	Learner progress dashboard queries (FR-L-09)
ai_sessions	learner_id, timestamp	AI session history per learner (FR-AI-01, NFR-SEC-05)
competencies	subject_id, level_id	Competency listing by subject + level (FR-L-09)
offline_cache	learner_id, sync_status	Offline content bundle listing per learner (NFR-O-01)
schools	region, district	Regional analytics aggregation (FR-M-03)

6. Traceability to Stage C & SRS

Stage C Entity	Stage F Table(s)	Key Design Decision	SRS Ref
User (abstract)	users	Base table for joined inheritance; UUID PK; bcrypt hash column	FR-L-01, NFR-SEC-03
Learner	learners	Extension table; FK = PK back to users; school_id FK	FR-L series
Teacher	teachers	Extension table; subjects_taught as TEXT[] array	FR-T series
MoESOfficial	moes_officials	Extension table; access_level ENUM for governance tiers	FR-M series
Content (abstract)	content	Base table; content_type discriminator column added	FR-L-03, FR-T-02
Lesson	lessons	Extension table; body_html column for offline rendering	FR-L-03, FR-L-04
TeacherMaterial	teacher_materials	Extension table; file_url for S3 reference	FR-T-02, FR-T-03
Subject	subjects	Standalone reference table; subject_name UQ	FR-L-03
ClassLevel	class_levels	Standalone; ENUM + sort_order for ordered display	FR-L-03
Competency	competencies	FK to both subjects and class_levels; composition enforced	FR-L-09
Post	posts	visibility ENUM default PRIVATE; is_deleted soft flag	FR-L-06, NFR-SEC-05
Comment	comments	FK to both posts and learners; append-only	FR-L-07
Reaction	reactions	Composite UQ(learner_id, post_id, type) prevents duplicates	FR-L-07
CollaborativeSession	collaborative_sessions	M:M with learners via session_participants junction	FR-L-08
CompetencyProgress	competency_progress	UQ(learner_id, competency_id) ensures one record per pair	FR-L-09
AISession	ai_sessions	kb_version column added for audit trail per query	FR-AI-01 to FR-AI-05
OfflineCache	offline_cache	sync_status ENUM for PWA sync state management	FR-L-04, NFR-O-01

Stage C Entity	Stage F Table(s)	Key Design Decision	SRS Ref
KnowledgeBase	knowledge_base	is_active flag; mcp_endpoint stored for AI Tutor routing	FR-AI-03
Report	reports	filters as JSONB for flexible query parameters; file_url for S3	FR-M-04
(Implied: school)	schools	Supporting table for school registration; region + district for analytics	FR-M-03
(M:M: Learner ↔ Session)	session_participants	Junction table; composite PK enforces uniqueness	FR-L-08
(M:M: Content ↔ Competency)	content_competencies	Junction table; replaces List<UUID> competencyIds from Stage C	FR-L-03, FR-L-09

7. Document Sign-off

This Database & Data Design document represents the agreed baseline schema for the CBC Digital Learning Platform following completion of Stage F. All implementation of the PostgreSQL schema on AWS RDS shall conform to the tables, columns, data types, constraints, and index specifications defined in this document.

Stage G (UI/UX Design) and Stage H (Architecture Validation & Review) shall proceed with reference to this schema as the authoritative data layer specification.

Role	Name	Signature / Date
Project Supervisor	_____	_____
Software Architect / Group Leader	_____	_____
Database Designer	_____	_____